

YOLO-Assisted Deep Learning for Australian Raptor Identification with Accessible Citizen-Science Interfaces

Master's Research Manuscript | Release v1.5

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System: YOLO-assisted localisation + EfficientNetB4 classification

Dataset: 1,992 processed images, eight Australian raptor species

Evaluation: 206-image held-out test split, seed 42

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Abstract

This thesis presents a research prototype for Australian raptor identification under citizen-science image conditions. The system combines YOLO-based bird localisation with an EfficientNetB4 classifier trained on eight Australian raptor species from iNaturalist Australia and the Atlas of Living Australia.

The prototype implements a multilingual Flask interface, top-3 species alternatives, Darwin Core export, feedback logging and provisional AUSLAN motion illustrations. On a 206-image held-out test split, the classifier reaches 84.95% top-1 accuracy and 0.8482 macro-F1. Bootstrap confidence intervals, calibration analysis, family-level error analysis and Grad-CAM diagnostics are reported to support reproducibility and scientific caution.

The v1.5 release is positioned as an academic baseline rather than a production wildlife-identification authority. It documents the limits of the dataset, the provisional status of accessibility assets and a v2.0 roadmap for 14 species, detector fine-tuning and stronger split governance.

Chapter 1. Introduction

Australian raptors are ecologically important apex and meso-predators, but reliable species identification remains difficult for non-specialists. Citizen-science platforms can expand monitoring coverage, yet visual identification, biodiversity data interoperability and accessibility for Deaf participants remain open problems.

This thesis asks whether a compact and reproducible computer-vision pipeline can support Australian raptor identification while also exposing records in Darwin Core format and documenting a pathway for participatory AUSLAN validation.

Research Questions

1. Can a YOLO + EfficientNetB4 pipeline identify eight Australian raptor species with useful accuracy under citizen-science image conditions?
2. Can the prediction workflow produce auditable per-image predictions and Darwin Core-compatible records?
3. How should provisional AUSLAN signs be framed so they invite participatory validation without claiming authority?

Contributions

- A validated eight-species v1.5 raptor classification baseline.
- A YOLO-assisted localisation path for images and video frames.
- Per-image evaluation outputs in results/test_predictions.csv.
- Bootstrap, calibration and taxonomy-aware error analyses.
- A multilingual Flask app with feedback and Darwin Core export.
- A documented AUSLAN consultation protocol.

Chapter 2. Literature Review

The project draws on four connected bodies of work: fine-grained bird classification and transfer learning; object detection for ecological image analysis; citizen-science biodiversity infrastructure, including ALA, GBIF and Darwin Core; and accessibility research, AUSLAN and participatory design.

EfficientNetB4 is used as the single v1.5 classifier because it offers a strong accuracy-to-compute balance at 380 pixel inputs. YOLO is used for localisation and candidate cropping, but the current release uses an adaptive policy because crop-only inference did not improve the held-out checkpoint.

The accessibility contribution is deliberately framed as a participatory prototype. The project does not assert that generated motion illustrations are validated AUSLAN signs; instead, it documents a consultation protocol that can be reviewed by Deaf community participants and interpreters.

Chapter 3. Methodology

Dataset

The v1.5 processed dataset contains 1,992 images across eight species. The active class order is fixed in `gui/app.py::CLASS_ORDER` and is verified by automated tests so that the model, dataset and user interface remain synchronized.

Split	Images
Train	1,590
Validation	196
Test	206

Source: processed project dataset, deterministic seed 42.

Class key	Common name	Family
<code>aquila_audax</code>	Wedge-tailed Eagle	Accipitridae
<code>circus_assimilis</code>	Spotted Harrier	Accipitridae
<code>elanus_axillaris</code>	Black-shouldered Kite	Accipitridae
<code>falco_cenchroides</code>	Nankeen Kestrel	Falconidae
<code>falco_peregrinus</code>	Peregrine Falcon	Falconidae
<code>hieraaetus_morphnoides</code>	Little Eagle	Accipitridae
<code>lophoictinia_isura</code>	Square-tailed Kite	Accipitridae
<code>tachypiza_fasciata</code>	Brown Goshawk	Accipitridae

Source: `gui/app.py::CLASS_ORDER` and project taxonomy metadata.

Pipeline

4. YOLO detects bird regions in an uploaded image or sampled video frame.
5. EfficientNetB4 classifies the whole image and, when useful, the YOLO crop.
6. The adaptive policy uses the crop only when it is more confident than the whole-image prediction.
7. The interface displays top-1 confidence and top-3 alternatives.
8. User feedback and Darwin Core exports are written to CSV artefacts.

Reproducibility

Artefact	Purpose
requirements.txt	Python dependency lock for local and CI execution.
environment.yml	Conda-compatible environment definition.
Dockerfile	Containerized reproducibility path.
.github/workflows/ci.yml	Lightweight CI with real pytest and healthcheck.
notebooks/healthcheck.py	Project contract checks for release integrity.
tests/test_project_integrity.py	Automated tests for Flask, i18n, Darwin Core and YOLO wrapper.
docs/DATASHEET.md	Dataset provenance, risks and constraints.
docs/MODEL_CARD.md	Model behavior, metrics and intended use.

Chapter 4. Results

Metric	Value
Test images	217
Top-1 accuracy	0.8387
Macro-F1	0.8311
Weighted-F1	0.8377
Expected calibration error	0.0639
Family-level accuracy	0.9272

Source: results/reporte_final.json and results/calibration_efficientnet_b4.json.

Species	Precision	Recall	F1	Support
Wedge-tailed Eagle	0.8462	0.9706	0.9041	34
Spotted Harrier	0.8710	0.8438	0.8571	32
Black-shouldered Kite	0.7500	0.8889	0.8136	27
Nankeen Kestrel	0.8421	0.7273	0.7805	22
Peregrine Falcon	0.8500	0.7391	0.7907	23
Little Eagle	0.7727	0.7727	0.7727	22
Square-tailed Kite	0.8214	0.9200	0.8679	25
Brown Goshawk	0.9615	0.7812	0.8621	32

Source: held-out test predictions for EfficientNetB4.

Metric	Bootstrap mean	95% CI
Accuracy	0.8504	[0.8010, 0.8981]
Macro-F1	0.8467	[0.7964, 0.8935]

Source: results/bootstrap_ci_efficientnet_b4.json, n = 1,000 bootstrap resamples.

The model reaches 92.72% family-level accuracy. Cross-family errors account for 15 of 206 test images (7.3%). The most common reported confusion is tachyspiza_fasciata predicted as elanus_axillaris, followed by falco_cenchroides predicted as elanus_axillaris.

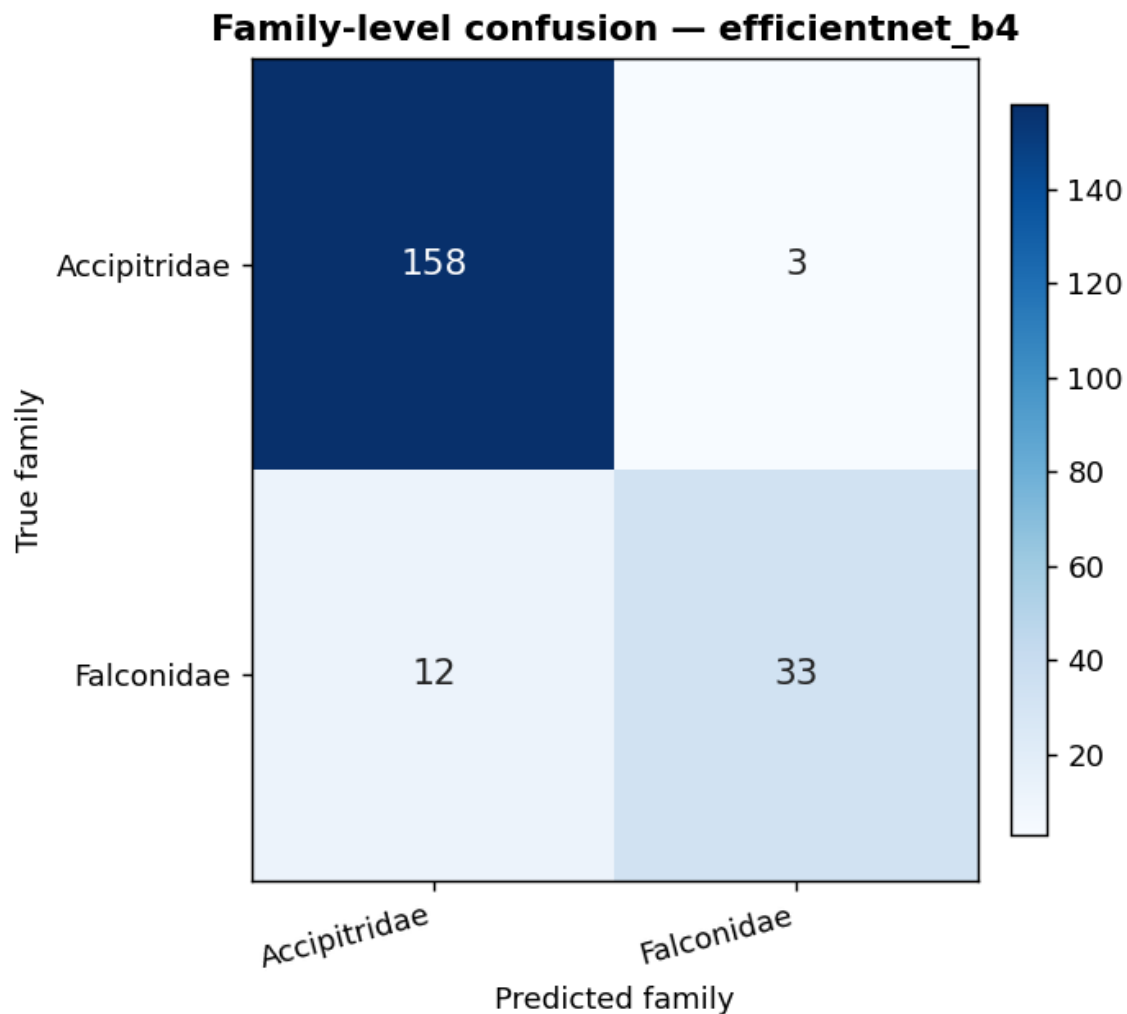


Figure 1. Family-level confusion matrix for the v1.5 EfficientNetB4 classifier.

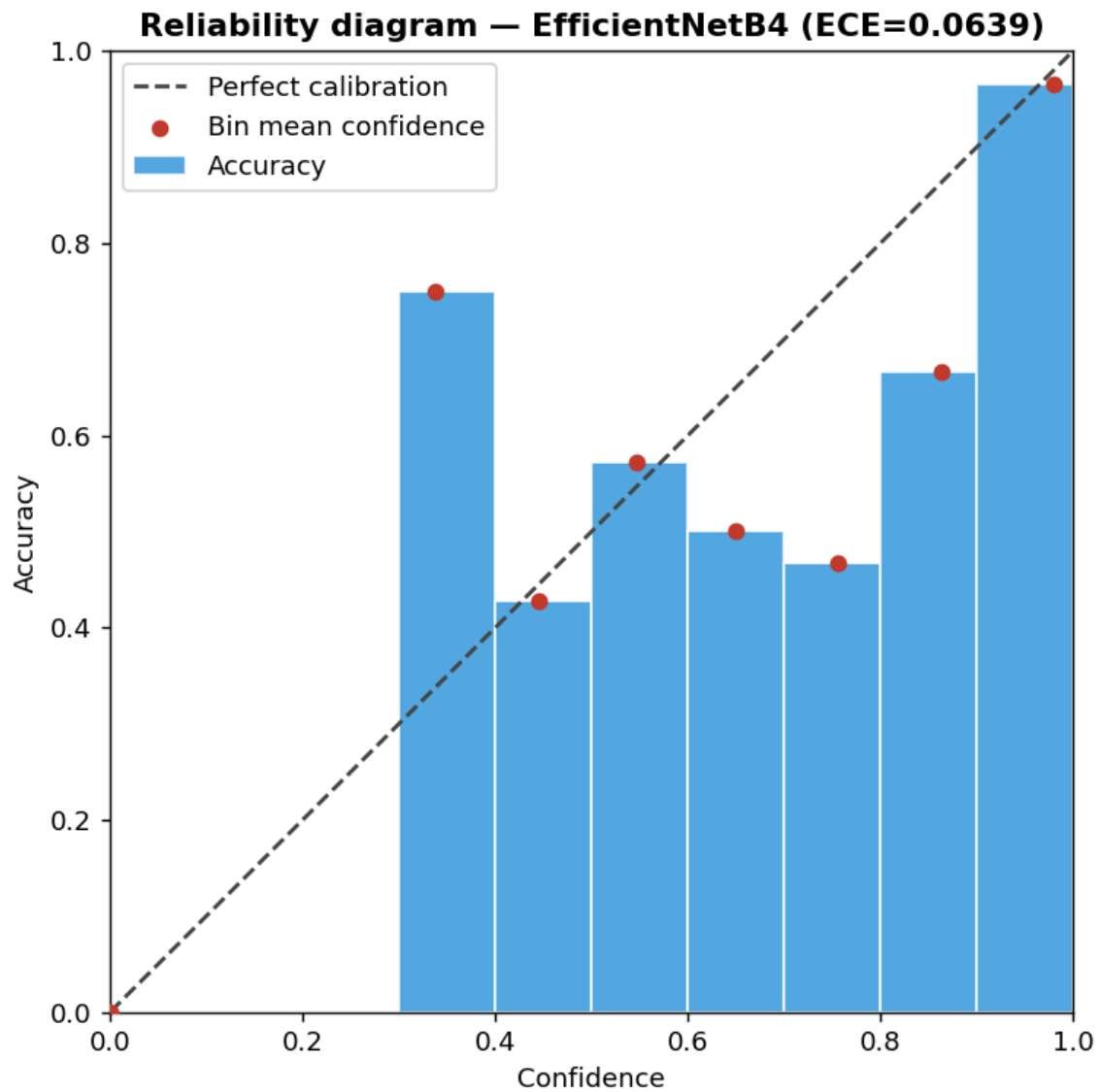


Figure 2. Reliability diagram used to estimate expected calibration error.

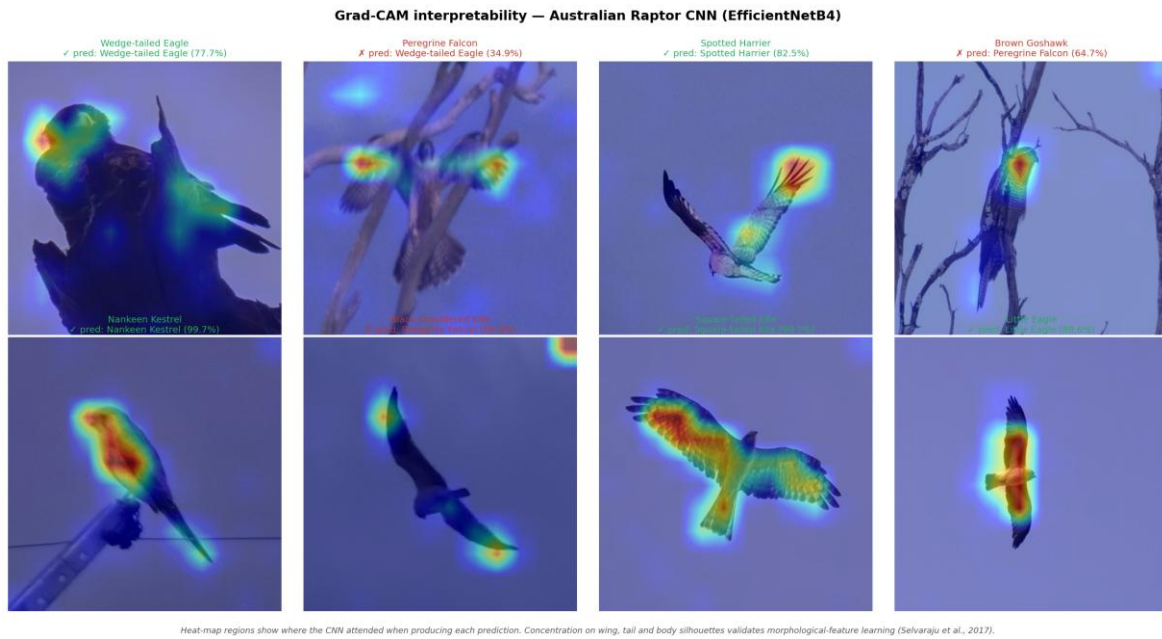


Figure 3. Grad-CAM mosaic showing image regions influencing classifier decisions.

Chapter 5. Software, Accessibility and Data Interoperability

The Flask application operationalizes the model as a citizen-science prototype. It supports image upload, video-frame sampling, top-3 alternatives, feedback logging, out-of-domain feedback, Darwin Core export and 10-language user-interface text.

System area	Implemented evidence
Flask routes	Health, identify, feedback, Darwin Core and static interface routes are covered by pytest.
Prediction audit	results/test_predictions.csv stores image_path, y_true, y_pred, confidence and top3.
Darwin Core	Export fields include scientificName, taxonID, basisOfRecord and identifiedBy.
Feedback	Incorrect predictions and out-of-domain reports are appended to CSV logs.
i18n	Ten translation files are checked for active species coverage.
YOLO	gui/yolo_detector.py provides optional Ultralytics-based bird localisation.

The AUSLAN component is explicitly provisional. No sign is claimed as validated AUSLAN until reviewed through Deaf-community consultation. The contribution is therefore a process and prototype contribution, not an authoritative vocabulary.

Chapter 6. Discussion and Conclusion

The v1.5 release demonstrates that a modest, reproducible YOLO-assisted EfficientNetB4 pipeline can support Australian raptor identification with useful but imperfect performance. The system is strong enough for a research prototype and scholarship portfolio, but not for legally consequential deployment.

Limitations

- The split is per-image rather than group-aware by photographer, location or observation event.
- The release is closed-set and limited to eight active species.
- Juvenile plumages, difficult angles and remote regions remain under-represented.
- AUSLAN illustrations are provisional and require community validation.
- YOLO uses generic bird/person-era detector behaviour rather than raptor-specific fine-tuning.

Future Work

- Expand v2.0 to the 14-species dataset after provenance and split validation.
- Keep EfficientNetB4 as the primary classifier while improving split governance, detector tuning and calibration.
- Fine-tune YOLO on raptor bounding boxes and report detector metrics.
- Introduce group-aware splits and geographic bias analysis.
- Run participatory AUSLAN validation before claiming signed vocabulary support.

The central conclusion is that the project is now defensible as a transparent academic baseline: it contains real metrics, real automated tests, explicit limitations, reproducible artefacts and a clear pathway from Master's prototype to stronger future doctoral research.

Appendix A. Dataset Datasheet Summary

Field	v1.5 value
Sources	iNaturalist Australia and Atlas of Living Australia.
Species	Eight active raptor species.
Images	1,992 processed images.
Splits	Train 1,590; validation 196; test 206.
Risks	Geographic, photographer, seasonal and plumage bias.
Standard	Darwin Core export supported for prediction records.

Full datasheet: docs/DATASHEET.md.

Appendix B. Model Card Summary

Field	v1.5 value
Classifier	EfficientNetB4, eight output classes.
Detector	YOLO localisation with an adaptive crop policy before classification.
Intended use	Research prototype and citizen-science assistance.
Out-of-scope use	Legal, veterinary, conservation enforcement or final expert identification.
Performance	Accuracy 0.8495; macro-F1 0.8482; app-output ECE 0.0639.
Known weakness	Closed-set errors, juvenile plumage and cross-family confusions.

Full model card: docs/MODEL_CARD.md.

Appendix C. Reproducibility Checklist

- Run `python notebooks/export_test_predictions.py` to regenerate per-image predictions.
- Run `python notebooks/bootstrap_metrics.py --report-md` for confidence intervals.
- Run `python notebooks/error_analysis.py` for family-aware error analysis.
- Run `python notebooks/calibration_ece.py` for ECE and reliability diagrams.
- Run `python notebooks/run_tests.py` for automated project tests.
- Run `python notebooks/healthcheck.py --verbose` for release-gate checks.

Appendix D. Defence Demonstration Script

9. Open the README and state the v1.5 scope: eight species, EfficientNetB4 and YOLO adaptive localisation.
10. Show `results/test_predictions.csv` and explain `image_path`, `y_true`, `y_pred`, confidence and top3.
11. Show the bootstrap, calibration and error-analysis artefacts in `results/`.
12. Run the test suite and healthcheck to demonstrate reproducibility.
13. Open the Flask interface and perform one image prediction.
14. Export a Darwin Core row and show how feedback is logged.
15. Close with limitations and the v2.0 research roadmap.

Appendix E. Release Package

Artefact	Purpose
<code>docs/Australian_Raptor_Thesis_v1_5.docx</code>	Editable thesis manuscript.
<code>docs/Australian_Raptor_Thesis_v1_5.pdf</code>	PDF thesis export for review and submission packets.
<code>results/thesis_docx_audit.json</code>	Structural DOCX audit for headings, tables, figures and metrics.
<code>results/thesis_pdf_audit.json</code>	PDF audit for page count, text extraction and required terms.
<code>RELEASE_MANIFEST_v1_5.md</code>	SHA-256 manifest for release source, documentation and result artefacts.

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